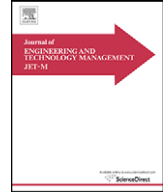




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The social structure of leadership and creativity in engineering design teams: An empirical analysis

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ABSTRACT

Creativity is essential for research and development efforts. Unfortunately, little is known about how the role of team leaders determines the team's creativity. Based on a sample of 39 engineering design teams in the space industry, this study examines the effects of leader position within different flows of communication on team creativity. The results indicate that the balance between holding a central or a peripheral position indeed determines the creativity of such teams. Specifically, very central and very peripheral positions of team leaders within the work-flow and awareness network hamper team creativity, whereas peripheral situated team leaders within the information network propel the creativity. In addition, team leaders stimulate creativity when they stay central in the external information network. In managerial terms, the analyses implicate that stimulating engineering design team creativity requires team leaders to smartly limit their involvement in communication and mainly act as gatekeepers to external sources of information.

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1. Introduction

As firms struggle to establish or maintain prosperity in turbulent and competitive environments, creativity becomes increasingly important (DeVanna and Tichy, 1990). Researchers widely agree that creativity is essential for the long-term survival of organizations, since it is the foundation of organizations' innovative potential (Badawy, 1993; Glynn, 1996; Oldham and Cummings, 1996; Scott

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and Bruce, 1994; Shalley, 1995; Unsworth, 2001). Indeed, there are very few business functions for which a fair level of creativity is not important. A marketing department is highly dependent on creative thought to generate ways to catch and hold the attention of (potential) customers. In product development, much creativity is required in finding novel answers to customers' wants and needs and in discovering technical solutions to bring them into being. Similarly, manufacturing demands creative approaches in devising (more) efficient ways of manufacturing. Distinctive about post-industrial society is the emergence of categories of work in which creativity and intellectual skills are distinguishing features. Creativity is now often at the basis of a company's competitive position since it is the design, development, marketing, and selling of products and services rather than their manufacture which are at the core of the business (Davis and Scase, 2000).

Accordingly, increasing attention has been paid to understanding creativity, especially in the last decade (Cummings and Oldham, 1997; Ford, 1996; Kratzer et al., 2004; Leenders et al., 2003; Woodman et al., 1993). Like other crucial organizational outcomes, the creativity of engineering design teams stems not only from overall firm strategy and access to resources but, more fundamentally, from the minds of the individual employees who, with others, carry out the work of the organization on a daily basis (Amabile et al., 2004). The extent to which they will produce creative – novel and useful – ideas during their everyday work depends not only on their individual characteristics, but also on the work environment around them (Amabile et al., 2004; Perry-Smith and Shalley, 2003; Reiter-Palmon and Illies, 2004). In the contemporary knowledge-work-intensive environments that engineering design teams face, most projects are performed by teams of professionals striving to be creative in developing new products, new services or new processes (Griffin, 1997; Van Engelen et al., 2001). Of all of the forces that impinge on people's daily experience of the work environment in teams, one of the most immediate and potent is likely to be the social process in which the team members are embedded (Leenders et al., 2003; Perry-Smith and Shalley, 2003). In all three prominent creativity models (Amabile, 1988; Ford, 1996; Woodman et al., 1993), the social processes are important antecedents to creativity.

This more social view of creativity is supported by a limited but growing collection of empirical evidence. Since the core product of creativity is new knowledge and new knowledge can only be created when existing bases of knowledge are disseminated through interaction between specialists with varying areas of expertise, creativity is couched in communication networks. The association between knowledge transfer and communication networks is widely confirmed in the scientific literature (Hansen, 1999; Ingram and Roberts, 2000; Patrashkova and McComb, 2004; Reagans and Zuckerman, 2001; Uzzi, 1996; Tsai, 2001). Communication is the means through which a team stores and disseminates information and more frequent communication may make the cross-fertilization of ideas increasingly likely (Mumford and Gustafson, 1988), but on the other hand when communicating too much the cognitive maps of the team members become too similar, and the group is assumed to be incapable of creative outputs (Patrashkova and McComb, 2004).

The creativity of engineering design teams is, therefore, largely shaped by the communication coordinated among the members of the team (Leenders et al., 2003). Leadership of these teams, or those "local leaders" who coordinate the member's access to knowledge and information, balance these communication networks (Amabile et al., 2004). Surprisingly, the balancing role of leadership has not generally been treated as a particularly important influence on creativity (Mumford et al., 2002), despite the likely impact of leadership on the work environment (Shalley and Gilson, 2004) and the demonstrated impact of the work environment on creativity (Amabile, 1996; Leenders et al., 2003; Mumford et al., 2002). Moreover, with the emergence of more matrix-like organizations the lightweight team leader role has continuously been redefined toward one of more importance (Norrgrén and Schaller, 1999). In particular, engineering design teams are increasingly autonomous units in which leadership is a decisive factor (e.g., Aronson et al., 2006).

In this study we will focus on the position of team leaders within different team-internal communication networks and their external communication, as well as on the creativity of engineering design teams. Our main goal is to create applicable knowledge about how the leader position in different internal communication networks and in their external communication affects the creativity of engineering design teams. In order to explore this issue we have gathered data on 39

engineering design teams engaged in the space industry around the globe. In the first part of the paper we will review the existing literature on leadership and creativity and clarify our concepts of leader position and communication networks. In this part we will introduce the balance between a peripheral and central position for team leaders as the most important positional characteristic, and then distinguish three team-internal communication networks, the workflow network, the problem-solving network, and the awareness network. In addition, we will introduce the boundary-spanning capacity of the team leaders, defined as their engagement in external communication. Thereafter, we will formulate hypotheses about the relationship between the leader's position in different team-internal communication networks, their boundary-spanning capacity and engineering design team creativity. In the second part of the paper we will describe our empirical study and present the results. Finally, we will discuss the results, draw academic and practical conclusions, and build applicable knowledge on how to lead engineering design teams toward creativity based upon these conclusions. The central implication of our findings is that, in order to navigate engineering design teams toward creativity, leaders have to smartly limit their communicational involvement within internal communication networks and concentrate on the role of gatekeepers linking the external and internal communication networks.

2. Relevant literature

2.1. *The balance between peripheral and central leader positions and creativity*

The literature linking leadership to group performance is scant (Kim and Yukl, 1995; Amabile et al., 2004), and the literature investigating the impact of leadership on creativity is even scantier. A handful of prior studies have found positive links between creativity and some specific team leader behaviors, such as serving as an ambassador for the team within the organization (Ancona and Caldwell, 1992), focusing on work-related rather than administrative communications (Katz and Tushman, 1979), providing various forms of assistance and support to the team's work (Amabile et al., 2004; Madjar et al., 2002; Shalley and Gilson, 2004; Tierney and Farmer, 2004; Zhou and George, 2003), and providing general supervision (Barnowe, 1975). According to leader-member exchange theory (e.g., Grean et al., 1982; Gerstner and Day, 1997), each employee establishes a social exchange relationship with his or her supervisors, and the quality of this leader-member exchange is generally found to be positively related to job performance (Janssen and Van Yperen, 2004). Put somewhat differently, leaders can offer valuable input to team processes and coordinate the dissemination of this input to each single team member (Day et al., 2004). In this way team leaders link the external and the internal communication networks. Both the external and internal communication networks and their structures are recognized as extremely critical for innovation endeavors (e.g., Chen, 2005; O'Connor and McDermott, 2004). When these networks are critical to innovation outcomes the position of the leader then is inherently critical also, since team leaders are by definition the most important network players.

In the network literature a number of different network positions are described and conceptualized. The balance between a central position and a peripheral position is among the most fundamental concepts.

When looking at the communication network in Fig. 1, the position of leader A is highly central since he/she is involved in many (strong) communicational linkages with his/her subordinates, whereas leader position B without any communicational involvement is mostly peripheral. Dating back to the MIT studies conducted by Bavelas (1950) and Leavitt (1951), researchers have found the balance between peripheral and central network positions to be associated with social power and structural influence. Mullen and Johnson (1991) conclude that holding a central position is a significant and independent predictor of leadership. In this study, therefore, we will apply the balance between peripheral and central positions since it focuses on social power and structural influence, which gradually arises from holding a more central position in communication networks (Bacharach and Lawler, 1980).

Research investigating intra-organizational social networks has found that central network positions and the opportunity to connect effectively to others in the organization provide access to

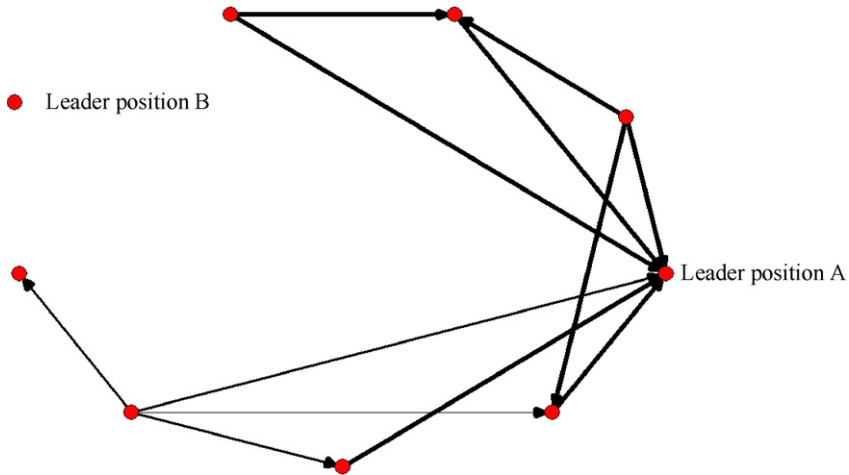


Fig. 1. Central and peripheral positions of engineering design team leaders.

knowledge and information and improve the ability to innovate (Hansen, 2002; Tsai, 2001). This research, however, concentrates on the business unit level and does not capture the intensity and richness of interpersonal communication networks (Oliver and Liebeskind, 1998). In addition, most network-studies researchers have traditionally focused on the structure of networks (Burt, 1992; Granovetter, 1973; Sparrowe et al., 2001) and paid less attention to the theoretically important features of the ties, or communication linkages, within these networks (Adler and Kwon, 2002; Higgins and Kram, 2001; Monge and Contractor, 2001). In social science these linkages are thought of as a network with specific contents, in which different contents of linkages may yield different effects on individual and team outcomes (Berkowitz, 1982; Marsden, 1990; Wellman and Berkowitz, 1988). Likewise, network structure alone cannot capture the effects that such ties have on knowledge and information acquisition.

In earlier research in innovative settings, differing network contents have been distinguished. For example, in a study by Norling (1996), content networks, situational networks, and process or workflow networks are distinguished. In another study by Morelli et al. (1995) three types of communication were discerned: first, coordination type; second, knowledge type; and, finally, inspiration type. Smith-Doerr et al. (2004) distinguish instrumental, expressive, technical advice, and organizational advice networks in a study of narratives of R&D team members. All these different distinctions show similarities and mirror the information necessary to fulfill the tasks in engineering design teams.

First, there are networks that describe the exchange of certain goods, data, or services, in engineering design teams this would more often be blueprints, drawings, etc., which are labeled differently in workflow, coordination type or instrumental networks. The engineering design task inherently refers to the generation of data, which have to be understood in combined in a mutual context. Exchanging work-related data, therefore, is a precondition for executing tasks in engineering design teams. The second common network type describes the exchange of different information or knowledge, such as knowledge type, technical/organizational, and content networks. This type of network content mirrors the mutual search for problem solutions based on the workflow network contents. Since research and development requires interaction, team leaders have to make sure that this interaction takes place. The fulfillment of this task can be traced when investigating the problem-solving network of teams. Interaction may be a very time-consuming activity, therefore it is important to interact with the right person at the right time. And, third, another dimension attempts to capture the non-work-related side of communication in terms of situational, inspiration or expressive networks. In a very recent work by Cross and Cummings (2004), this type of network is labeled an awareness network, describing the network of knowledge that team members have about each other.

In particular, in a knowledge-intensive environment it is extremely important to know where the sources of expertise, knowledge and information are in order to complete the design tasks.

All these three network types may be present in the workplace independently of each other. The work process, however, inherently creates a certain degree of overlap, since interaction within all three networks is essential in order to complete the tasks in engineering design teams. In addition, when taking a closer look at leadership tasks the derivation of these three network contents seems logical. The first major task of team leaders is the management of the work processes in teams. The execution of this task can be described in the workflow network. Second, team leaders have to make sure that the work process results in mutual problem-solutions mirrored in the problem-solving network. And third, in knowledge-intensive settings team leaders have to orchestrate the combination of different knowledge pools represented within the awareness network. Subsequently, in our study we will apply these three network types in terms of the workflow network, the problem-solving network, and the awareness network.

Besides investigating these team-internal networks, we will examine the leader position within the external network labeled as boundary-spanning capacity. The external network is much more broadly defined than the three internal networks described above. This network captures all work-related contacts with other engineering design teams, involving workflow, problem solving or awareness interaction. The boundary-spanning capacity describes the extent to which leaders of engineering design teams are in contact with other engineering design teams within the external network.

3. Research hypotheses

3.1. *The workflow network: leader position and creativity*

Engineering design team members have no choice but to have some kind of work-related exchange of output and input. In a recent study, Kratzer et al. (2006) indicate that the coordination of the workflow network can be used as a managerial tool to inspire creativity in innovation teams. There are two components that mainly determine this structure. First, there is the decomposition of the task at hand and the resulting interfaces that require team member exchanges. And, second, there is the coordinating hand of the team leader who can connect or disconnect team members for managerial reasons. In innovation trajectories this structure inevitably changes and adapts to the shifting challenges and tasks. The role of team leaders in workflow networks is twofold. Team leaders as coordinators monitor the workflow and change or intervene when the situation requires it. On the other hand, team leaders themselves have to be so involved in the workflow as to be able to evaluate the work progress and to navigate the team.

This twofold role has certain consequences for the creativity of the team. Team leaders who are strongly involved in the workflow network easily become overloaded with knowledge and decision needs, and this impedes creativity since team members may receive the necessary input delayed or not at all. In addition, a strong central position within the workflow network may discourage team members from finding novel solutions by centralizing the responsibilities of work-related exchanges. On the other hand, without the monitoring and navigating execution of the manager's work the team may shift away from seeking creative solutions. This phenomenon may occur when communication problems arise or certain members free-ride on the contributions of others. This development may go so far that the set standards or requirements are lowered by the team members themselves, resulting in poor creative solutions.

Hypothesis 1. Very peripheral and very central positions of team leaders within the workflow network impede the creativity of engineering design teams.

3.2. *Problem-solving network: leader position and creativity*

Creativity in knowledge-intensive environments is, to some degree, a product of obtaining the right knowledge and information at the right time to solve novel, challenging problems. High team-leader centrality in information sharing is commonly believed to inhibit creativity (Leenders et al., 2003).

Team leaders who dominate discussions and searches for innovative solutions are prone to information overload. The more innovative and complex the new product tasks, the more different bases of expertise are required. Especially in highly technological areas, no single team member, regardless of tenure, status or skill, can have enough knowledge to oversee all areas of expertise proficiently. The team leader will likely impede engineering design team creativity by wasting time and distorting the problem-solving process.

Team members may not receive, or receive too late, the information necessary for their part of the task. This may require them to discard previous work in light of this “new” information and start all over again. In addition, when communication is less centralized, accountability for the team’s results (both in terms of end results, intermediate results and milestones) becomes more evenly distributed across team members. Every member then becomes engaged in the process and in finding creative solutions to the problems at hand. Moreover, the presence of highly central team members reduces the autonomy of non-central individuals. This autonomy is instrumental in creative achievement (Amabile, 1983).

Hypothesis 2. The more central the position of the team leader within the problem-solving network the lower the creativity of engineering design teams is.

3.3. Awareness network: leader position and creativity

In knowledge-intensive work, workflow and problem-solving networks are often dynamic, and they shift when new projects demand different kinds of information and expertise. As new problems or opportunities arise, the set of relationships one is currently tapping for information can turn out to be less helpful than that from those other group members that one is not yet aware of, who might have information relevant to the new tasks at hand. A network can supplement a person’s ability to respond well to new challenges when that person knows whom to seek out for knowledge and information relevant to a new project. Thus, team leaders who are more aware of a team member’s expertise should be more likely to link together the right people at the right time when unique challenges present themselves.

According to Borgatti and Cross (2003), one’s awareness of another’s expertise, or the extent to which one person knows what another person knows, is associated with the likelihood of seeking information from that person. In an engineering design team, however, very specialized expertise is scattered within the team such that no member can possibly oversee all the areas of expertise of his or her colleagues. Team leaders, however, who are aware of the distribution of knowledge and expertise within the team, can coordinate the team in such a way that the right members at the right time exchange their knowledge and information. As Cross and Cummings (2004) indicate, team members with greater awareness of disparate expertise within a network improve their ability to respond appropriately when new challenges demand different knowledge.

Hypothesis 3. The more central the position of the team leader within the awareness network the higher the creativity of engineering design teams is.

3.4. Boundary-spanning capacity and creativity

As indicated earlier, team leaders have the ascribed role of gatekeepers who link the external and internal communication networks. Thus, leaders can remain central within the internal networks but also maintain a number of contacts with other engineering design teams in the same organization. The degree to which the team leaders are linked to other teams describes their external centrality or boundary-spanning capacity. Boundary spanning has long been known to influence how information enters organizations (Tushman, 1977; Tushman and Katz, 1980). For example, ties crossing boundaries are associated with adoption of less normative strategies (Geletkanycz and Hambrick, 1997), acquisition of competitive capabilities (McEvily and Zaheer, 1999), and product innovation (Hargadon and Sutton, 1997).

Early work demonstrated the critical role of gatekeepers crossing technical boundaries (Allen, 1977). More recently, research has demonstrated the importance of ties crossing departments or functional boundaries for effective knowledge transfer within organizations (Hansen, 2002; Szulanski, 1996; Tsai, 2001). Team leaders who remain central in the external communication network have a high boundary-spanning capacity since they can gather requested information easily in their network. Subsequently, a high boundary-spanning capacity enables team leaders to provide the engineering team members with the right information and knowledge at the right time.

Hypothesis 4. The more central the position of the team leader within the external communication network (or the higher the boundary-spanning capacity) the higher the creativity of engineering design teams is.

4. Methods

4.1. Study setting

To explore the social structure of leadership and creativity in engineering design teams, we collected data on 39 engineering design teams with 321 team members engaged in the space industry around the globe in 17 countries. All these engineering design teams are part of three large instrument consortia coordinated by the European Space Agency (ESA). All three consortia are engaged in the development of telescopes and are comparable in terms of budget, time-lines and complexity. Going into more detail, for all three consortia their scheduled budgets are higher than \$250,000,000; they have a scheduled tenure from 11 to 14 years; and all three face very high complexity by developing completely new technical solutions. The three large instrument consortia employ 59 engineering design teams. Many team leaders, however, are responsible for more than one engineering design team, so we selected the teams that have different team leaders to prevent theoretical as well as analytical problems.

In order to do so we randomly selected one team for each team leader responsible for more than one team. The teams mainly employ male members; the percentage of women is negligible with less than 3%. The teams are on average 4.5 years old and, with the exception of a very few cases, they have employed the same team members throughout this period of time. So, the social structure of the teams can be described as very stable. Most team members have a Master's degree (56%), 34% of the team members have a Bachelor's degree, and 10% a PhD.

For gathering the required data we used semi-structured interviews and questionnaires. All 39 team leaders were interviewed and completed a questionnaire resulting in a response rate of 100%. All team members were asked to fill in a questionnaire resulting in a response rate of 72%. The team leaders were approached during formal team meetings and the team member questionnaires were sent by mail. The individual identification for the interview and questionnaire data was removed once the data had been entered, such that all information was maintained but information about the identity of the individuals was invalidated. In this way the survey was kept completely anonymous. In all the engineering design teams, preliminary interviews with the project management, observation and existing documentation confirmed that the work in each setting was highly knowledge intensive and collaborative. Moreover, the data collected were widely confirmed by this additional information.

4.2. Defining and measuring the creativity of engineering design teams

There are many views of creativity. Amabile (1988) argues that creativity is exhibited when a product or service is generated that is both novel and useful with respect to the firm. Woodman et al. (1993, p. 293) contend that creativity refers to “the creation of a valuable, useful new product, service, idea, procedure, or process by individuals working together in a complex social system.” King and Anderson (1990, p. 82) proposed that creativity at the team level explicitly incorporated the interpersonal discussion among team members. In the present research, we will build on Woodman et al. (1993) and take King and Anderson's proposition as a central theme in our study.

Much of the research on creativity has been conducted in experimental settings in behavioral laboratories, often with students and children as subjects. Although some of these experiments are set up to mirror organizational situations (e.g., [Amabile, 1996](#); [Carson and Carson, 1993](#); [Hackman and Oldham, 1980](#)), they can hardly capture the creative processes that occur in real organizational groups where the project management structure is composed of multifunctional teams ([Drazin et al., 1999](#)) that are too large to be experimentally viable. Moreover, the engineering design process is characterized by long-term horizons. Creative solutions to sub-problems may not turn out to be viable after solutions to other sub-problems have been devised (which may occur only much later in the project), and the separation in time between the creative act and its evaluation is often long. In addition, the creative task of engineering design teams is typically more multifaceted and intricate than research settings allow for.

Measuring creativity, therefore, is notoriously difficult. In the lion's share of creativity studies, creativity is measured in either of two ways. First, by the performance of individuals or groups of individuals on standardized creativity tests—constructed and scored in a similar way as are tests of verbal and mathematical intelligence. Alternatively, a second group of measures entails the rating of actual products in response to open-ended instructions. Rating is then performed by outside experts. These measures, however, are not appropriate for our research. Since we had no outside experts available to assess the teams' creativity, we asked inside experts with — for most of them — long experience in the field: the team leaders and the members themselves. On a seven-point scale we asked the team leaders and members to rate the team's creativity using the following items:

1. How would you estimate the novelty and originality of the solutions your team finds to problems? (7-point scale from 1-not new to 7-very new).
2. How would you estimate the number of possible solutions your team develops to solve problems? (7-point scale from 1-not high to 7-very high).
3. How would you estimate the number of possible solutions your team takes into consideration in order to solve problems? (7-point scale from 1-not high to 7-very high).

In order to have one measure of creativity, we combined the three items into one scale of creativity reaching from 1 (very low) to 7 (very high). The reliability of this scale was tested and is satisfactorily high with Cronbach's alpha at 0.82.

The measure for team-level creativity thus derives from an assessment by its leaders and members; in this sense, it is a quasi self-report measure. Self-report measures are often criticized, mainly with the argument that some people are unable to report their performance accurately due to reasons of poor introspection (e.g., [Locke et al., 1988](#)). Since we asked team members and leaders to evaluate their team's creativity rather than their own individual creativity, the effect should be much smaller in our data than in a situation in which individuals are asked to assess their own individual creativity. Moreover, in his well-known study of engineering design team's success factors, [Cooper \(1981\)](#) achieved accuracy levels of over 80% by asking team members and leaders to assess their projects ex-post.

In addition, if groupthink were present, this might affect team-member assessment of a team's creativity. If this effect in a team were strong, team members should collectively report extra-high (positive groupthink) or extra-low (negative groupthink) achievement, while displaying a low variation between them. This effect was tested for by dividing the sample into "extreme" scores (first and fourth quartiles) and "regular" scores (second and third quartiles). Indeed, standard deviations are lower for "extreme" teams than for "regular" teams. This effect, however, is not statistically significant ($p = 0.65$). Similarly, a discriminant analysis could not outperform a random model in assigning teams to the "regular" or "extreme" group. In sum, if groupthink is present, its effect on the creativity measure is only very limited and unlikely to affect the empirical findings.

Finally, there is evidence that self-ratings correlate highly with more "objective" measures in those cases where anonymity is promised. In particular, [Heneman \(1974\)](#) found that self-report measures had less restriction in range and leniency than did the purportedly more objective supervisor ratings. He suggested that this reflected the fact that self-reports were obtained explicitly for research purposes and were not to be used for evaluatory or other organizational purposes. The same holds for

our measure, since we promised that the responses would not be shared with management. They can thus be expected to paint a reliable picture of actual team-level creativity.

4.3. Measuring the communication networks

In the survey, three full rosters of team members (including the team leaders) according to the three team-internal network types were used for each engineering design team. Each team leader and member was asked to report the communication with all other team members in these rosters. The result of these questions was three valued asymmetric matrixes for each engineering design team with values 0 = no (or less than monthly) communication, 1 = (at least) monthly communication and 2 = (at least) weekly communication. So, for each team-internal network type, workflow, problem solving and awareness, an asymmetric matrix of the frequency of communication between all the team members (including the team leader) was created. This procedure is typical of the way in which communication network data are collected (e.g., Leenders et al., 2003) and includes weak and strong ties according to Granovetter (1973).

For the external communication we used the same scale. The question was addressed to the team leaders and members by presenting them with a full roster of all involved engineering design teams within the relevant instrument consortia. The result of this question was ego-centered networks of external communication for each team member and the team leaders. The ego-centered networks described the number and the frequency of external communication with other engineering design teams within the same consortia.

The separate team-internal networks were addressed using the following questions:

On average, how often do you interact with other members within your team about the following aspects?

Workflow network	<i>The exchange of facts (e.g., designs, manufacturing drawings, models or test data).</i>
Problem-solving network	<i>The communication involving the discussion, innovation or evaluation of new ideas or approaches to technical problems; technical or scientific help or advice; mutual use as “sounding board” for ideas; distribution of scientific or technical information stemming from outside of the team.</i>
Awareness network	<i>The exchange of information about the persons in the organization (e.g., knowledge about who has what role or task in the project or who has the formal authority to make certain decisions).</i>

The external communication network was addressed using the following question:

On average, how often do you exchange information and knowledge with members of other engineering design teams within your instrument consortia?

4.4. Defining and measuring the balance of peripheral and central positions

Among the proposed measures of positional centrality, degree centrality is the simplest and most straightforward (Zemljic and Hlebec, 2005). Degree centrality is based on the number of units directly connected to the unit under scrutiny. The definition of actor centrality is that the most central actor must be the most active, in the sense that this actor has the largest number of ties to other actors in the network (Freeman, 1979). In this way it measures the balance between having a peripheral position or a low degree of centrality or having a central position or a high degree of centrality. The measure is focused on the level of communication activity for internal communication networks on the intra-team or individual level and for the external network on the intra-team or unit level. In a directed network there are two distinct degree centralities, that is, in-degree (choices received) and out-degree (choices made). The distinction between out-links and in-links is particularly important in some cases since their interpretation cannot be the same. In our study, however, there is no meaningful distinction between out-links and in-links since two

connected team members communicate regardless of which team member sends or which one receives information (Barnowe, 1975).

This assumption is supported by empirical evidence showing that two-way interaction is more frequent in engineering design teams than one-way information flows (Morelli et al., 1995). We also executed the analyses separately for the in-degrees and out-degrees and did not find any differences in the results. In addition the reciprocity of indications in all three asymmetric matrixes was higher than 0.70 and highly significant ($\alpha < 0.001$), so we decided to symmetrize the matrixes. Symmetrized matrices also increase the robustness of the data. The analyses presented are based on the symmetrized matrixes. We also considered other measures of centrality, that is, betweenness, degree (Freeman, 1979) or eigenvector (Bonacich, 1987), but chose degree for parsimony. Moreover, the position of the team leader did not change if alternative measures were used. For computing the degree centrality we used UCINET 5.1 (Borgatti et al., 1999).

For the external network no distinction between out-degree and in-degree network can be made, since we measured the ego-centered networks indicating only whether a team member or leader is in contact with other engineering design teams of the same consortia. So, being central for team leaders within the external network means having many and frequent contacts with the other engineering design teams of the same consortia.

4.5. Controls

There are numerous other characteristics of teams that have been or may be shown to influence team creativity. While it is not possible to include all these variables in this study, we did control for three variables that have been suggested as affecting team creativity.

4.5.1. Cross-nationality

Since all three consortia consist of teams that are dispersed across a number of countries, there may be country effects mediating the relationship between leader position and creativity. Since the analyses do not allow including all participating countries as dummy variables in our analyses, we attempted to cluster the teams into larger categories. The majority of the teams are situated in Europe, another group in Asia, and a few teams in North America. A hierarchical cluster analysis, however, including all relevant variables, does not create any sense-making clusters of teams. A descriptive analysis of the teams, however, revealed that more than half of the teams consisted of members from different countries, so that more than half of the teams were multinational. When distinguishing between teams with members from different countries and teams with members from the same country, an executed *t*-test shows that multinational teams provide a higher boundary-spanning capacity ($t = 2.56, p < 0.05$). We, therefore, decided not to take the country or continent where the teams were situated into account, but to distinguish between teams that were multinational and teams that were not multinational. Accordingly we included a dummy variable labeled “Cross-nationality” into our analyses with value 0 for teams that consist of members from the same country and value 1 for teams that consist of members from different countries.

4.5.2. The team member's boundary-spanning capacity

As earlier research suggests (e.g., Allen, 1977; Tushman, 1977), not only the team leaders but also the team members provide external contacts. Thus the leader's boundary-spanning capacity may only be a proxy for the team's external contacts. We, therefore, included a control variable labeled “Team member's boundary-spanning capacity” in our analysis. When including this variable in the analysis it became possible to separate the effect of the leader's boundary-spanning capacity and the team member's boundary-spanning capacity on creativity. The variable “Team member's boundary-spanning capacity” represents the average degree of external contacts of the team members for each team excluding the external contacts of the team leaders. An executed correlation showed that the “Team member's boundary-spanning capacity” and the “Leader's boundary-spanning capacity” correlated with $r = 0.54$ ($p < 0.01$).

4.5.3. Density

The team network as a whole may also affect the creativity of the teams. As research indicates (e.g., Leenders et al., 2003) the density of the engineering design teams may directly impact on their creativity and the centrality of the team leader may be only a proxy for team density. A correlation between the density scores and the team leader's centrality reveals that the correlation coefficients vary between $r = 0.61$ regarding the workflow network, $r = 0.65$ regarding the problem-solving network and $r = 0.63$ with respect to the awareness network. The results show that the leader centrality scores can only explain up to 42% of the variance of the team's densities. In addition the correlations do not give any indication about the joint effects on creativity. We, therefore, included both the team's densities and the leader centralities in our analyses. The variable of "team density" represents the average-degree centrality of team members, excluding the team leader's contacts.

4.6. Data analysis

In this research, we are interested in the factors that explain the relationship between leadership and creativity in engineering design teams. In engineering design team settings where the object of creativity is complex and requires skills from multiple bases of expertise, it is difficult to separate out individual from team-level contributions. Teams also go through stages that mirror the processes of individuals—that is, developing criteria, generating alternatives, modifying those alternatives, and amplifying and extending original ideas. Individuals and teams participate in creative processes in an iterative fashion. In the present article, we will follow the suggestion of Drazin et al. (1999) and Klein et al. (1994) and assume individuals act homogeneously within the teams as they engage in creative behavior.

In other words, we assume there is no need to incorporate individual-level variables, such as functional background, education and tenure, into our analysis of team-level creative performance. We therefore will limit ourselves to incorporating team-level aggregates into our analysis. In order to justify this aggregation statistically, a one-way analysis of variance was conducted to determine if there was greater variability in team creativity ratings between teams than within teams. The analysis of variance supported the appropriateness of the aggregation ($F = 3.13, p < 0.000$). Prior to aggregating team members' evaluations, inter-rater agreement on the team's creativity was calculated by averaging the inter-rater reliability (IRR) score suggested by James et al. (1984). The IRR was 0.72, justifying the use of the arithmetic mean as a team score.

In order to test the hypotheses we executed multiple regressions. As a first step we analyzed the main effects of leader centrality in the different networks and creativity including the controls in Models 1–4. Thereafter, we analyzed the interaction of the hypothesized main effects by entering the independent variables into the regression Model 5. In addition, we explored non-linear features of the independent variable 'Leader centrality awareness network' in Model 6. Since the sample size is quite small with 39 teams, we did not include the controls in the additional analyses in Models 5 and 6. All models were tested for violations of multicollinearity by checking the VIF (variable inflation factor) and CI (condition index) and the distribution of residuals. These examinations did not reveal any violation. Testing Hypothesis 1 requires a polynomial effect to be included in the regression. However, the high correlation between the square of leader centrality in the workflow network itself would introduce problems of multicollinearity into the analysis. We solved this by centering leader centrality in the workflow network by subtracting the overall mean from it. The squared value incorporated into the regression analyses is the squared value of this centered variable.

5. Results

As Table 1 shows, the densities of the three team-internal networks positively correlate with each other and are statistically significant. These correlations are common, since in the process of work we expect the three networks to overlap. Still, the overlap is only medium-high with the highest correlation being $r = 0.49$. These results indicate that the three team-internal networks represent three distinguishable communication networks. The density of the problem-solving and awareness networks have a negative impact on the team's creativity at a statistically significant level. In addition,

Table 1Descriptives and correlations: controls ($n = 39$).

Variables	SD	Mean	1	2	3	4	5	6
1. Creativity	0.97	5.34	–	0.06	–0.26 [†]	–0.24 [†]	0.17	0.09
2. Density workflow network	0.34	0.05	–	–	0.41 ^{**}	0.37 ⁺	0.12	–0.11
3. Density problem-solving network	0.37	0.11	–	–	–	0.49 ^{**}	0.13	–0.05
4. Density awareness network	0.41	0.19	–	–	–	–	0.07	–0.06
5. Team member's boundary-spanning capacity	0.27	0.39	–	–	–	–	–	0.25 [†]
6. Cross-nationality	0.51	0.54	–	–	–	–	–	–

⁺ $p < 0.05$.^{**} $p < 0.01$.[†] $p < 0.10$.

engineering design teams with members from different countries provide a higher boundary-spanning capacity at a statistically significant level.

Table 2 shows that the distribution of involvement of the team leaders in the different team-internal communication networks varies. Team leaders spend most of their time communicating about problem-solving issues, followed by involvement in the workflow network, as the mean values indicate. The least amount of team leaders' time is devoted to being active in the awareness network. This is an interesting result because it shows how much leaders are involved in work-related activities. A second immediate result is that the correlations between the three team-internal networks are not higher than 0.41. This again indicates that we have indeed measured three distinct networks. The leader centrality in the workflow network and the leader's boundary-spanning capacity positively impacts on creativity at a statistically significant level. The leader centrality in the problem-solving network and the squared-leader centrality in the workflow network both have a statistically significant negative effect on creativity.

In Table 3 the main effects of team creativity and leader centrality were tested, including the control variables. In Model 1 it is shown that the relationship between leader centrality in the workflow network and team creativity is statistically significant and inversely U-shaped. In Model 2 the relationship between leader centrality in the problem-solving network and team creativity turns out to be negative at a statistically significant level. In Model 3 no statistically significant effect between leader centrality in the awareness network and team creativity could be detected. And finally Model 4 illustrates that there is a positive relationship between the boundary-spanning capacity of the team leader and team creativity at a statistically significant level. The densities of the problem-solving and awareness networks negatively impact on team creativity and are statistically significant in Models 2 and 3. The two control variables, team member's boundary-spanning capacity and cross-nationality, do not have any statistically significant effects. The results confirm Hypotheses 1, 2 and 4. Leader centrality in the awareness network again does not have any significant effect on creativity.

In Table 4 possible interaction effects from leader centrality in the three team-internal networks and the boundary-spanning capacity of the leader and team creativity were tested. Models 5

Table 2Descriptives and correlations: main variables ($n = 39$).

Variables	SD	Mean	1	2	3	4	5	6
1. Creativity	0.97	5.34	–	0.41 ^{**}	–0.22 [†]	–0.53 ^{**}	0.11	0.39 ^{**}
2. Leader centrality workflow network	0.56	0.31	–	–	0.11	0.05	0.41 ^{**}	0.34 ⁺
3. Leader centrality workflow network ²	0.23	0.09	–	–	–	0.15	0.19	0.12
4. Leader centrality problem-solving network	0.18	0.67	–	–	–	–	0.28 [†]	0.09
5. Leader centrality awareness network	0.30	0.19	–	–	–	–	–	0.43 ^{**}
6. Leader boundary-spanning capacity	0.33	0.47	–	–	–	–	–	–

⁺ $p < 0.05$.^{**} $p < 0.01$.[†] $p < 0.10$.

Table 3Regression results for team creativity ($n = 39$).

Variables	Model 1	Model 2	Model 3	Model 4
Intercept	5.74**	5.15**	4.34**	5.89**
Leader centrality workflow network	3.17** (0.53)			
Leader centrality workflow network ²	−3.55** (1.08)			
Density workflow network	0.78 (0.43)			
Leader centrality problem-solving network		−3.56** (0.91)		
Density problem-solving network		−1.45† (1.04)		
Leader centrality awareness network			−0.56 (0.34)	
Density awareness network			−1.09† (0.88)	
Leader boundary-spanning capacity				4.76** (1.45)
Team member's boundary-spanning capacity				0.83 (0.44)
Cross-nationality	0.56 (0.26)	0.64 (0.22)	0.71 (0.25)	0.75 (0.34)
Adjusted R^2	0.21	0.22	0.11	0.29

Unstandardized coefficients are shown (standard errors in parentheses). Two-tailed tests are reported.

* $p < 0.05$.** $p < 0.01$.† $p < 0.10$.

confirmed the results shown in Table 3 and implies that leader centrality within the workflow, problem-solving, and leader's boundary-spanning capacity do affect team creativity independently of each other. The statistically significant explanation of variance 45% in Model 5.

Models 5 also shows that Hypothesis 3 cannot be confirmed. It may be that team members are limited in their autonomy, when the team leader is too central. In addition, the process of searching for information may be a creativity-stimulating process on its own. In order to explore statistically these thoughts indicative of a quadratic relationship, we entered into the regression the quadratic term for leader centrality within the awareness network. This quadratic variable was then also centered by subtracting the overall mean from it so as to avoid problems of multicollinearity. The results are displayed in Table 4 Model 6. The regression Model 6 justifies this exploration since the relationship between the centrality of the team leader within the awareness network and creativity is indeed statistically significant and inversely U-shaped. This statistical significance, however, only reaches the 10% level when one runs the risk of making a mistake of Type I. The regression model's statistical significance is improved when one enters the quadratic term of leader centrality within the awareness network, and this explains 49% of the variance in creativity.

To summarize these results, it can, in fact, be stated that the centrality of the team leader in different networks does impact on the creativity of engineering design teams. To be more specific, the results confirmed Hypothesis 1 by showing an inversely U-shaped relationship between leader

Table 4Regression results for team creativity ($n = 39$).

Variables	Model 5	Model 6
Intercept	4.61** (0.47)	4.96** (0.49)
Leader centrality workflow network	1.89** (0.37)	1.88** (0.36)
Leader centrality workflow network ²	−3.55** (1.17)	−2.87** (0.78)
Leader centrality problem-solving network	−2.21** (0.66)	−1.93** (0.61)
Leader centrality awareness network	−1.19 (0.75)	1.12† (0.69)
Leader centrality awareness network ²		−3.06† (1.75)
Leader boundary-spanning capacity	4.56** (0.81)	4.34** (0.77)
Adjusted R^2	0.45	0.49
Significance of R^2 change		0.000

Unstandardized coefficients are shown (standard errors in parentheses). Two-tailed tests are reported.

* $p < 0.05$.** $p < 0.01$.† $p < 0.10$.

centrality in the workflow network and creativity. Also, the proposed relationship between the negative effect of leader centrality within the problem-solving network and creativity in [Hypothesis 2](#) has been confirmed. [Hypothesis 3](#), which proposed a positive relationship between leader centrality within the awareness network and engineering design team creativity, was not confirmed. In contrast, when testing this relationship an inversely U-shaped relationship turned up. The boundary-spanning capacity of the team leader strongly positive relates to the engineering design team's creativity, confirming [Hypothesis 4](#). In addition, the densities of the problem-solving and awareness networks negatively impacted on team creativity.

6. Discussion and implications

Throughout the last decades, creativity has been recognized as one of the most important factors ensuring prosperity for a company's existence. Perhaps the "creating-creative" organization of the future will be one that integrates and moves beyond the promises of models for the new organization in the third millennium, such as the learning organization ([Sage, 1990](#)) or the knowledge-generating company ([Nonaka and Takeuchi, 1995](#)). Creativity has long been studied on the individual level by looking at certain personal traits that generate or impede creativity. Since the pioneering work of [Amabile \(1988\)](#), [Ford \(1996\)](#), and [Woodman et al. \(1993\)](#), we know that creativity is more than an individual characteristic.

The research investigating creativity has increasingly come up against factors in the environment of individuals that stimulated or discouraged creativity. The most often-cited factor has been the social process in which individuals are embedded. Social processes can be studied and mapped by looking at communication networks. Since communication networks exemplify the flow of knowledge and information, they are one of the most important objects to study in order to unravel the secrets of creativity. In the communication networks of teams, the team leader has by ascription the most important position. In particular, engineering design teams work in an increasingly autonomous way so that the team leader has become even more vital to the functioning of the group.

In the 1960s, team leaders rarely had autonomous responsibilities. This situation has changed through time so that today's team leaders, especially in larger projects, have the majority of the responsibilities and report directly to higher management. It is surprising that hardly anyone has investigated the leader's position within the different communication networks and its impact on the creativity of teams. It is exactly this issue that has been investigated in engineering design teams who are at the forefront of creative challenges. To be specific, we examined the intra-team centrality of leaders in three team-internal networks, the workflow network, the problem-solving network, and the awareness network. In addition, we looked at the external activity of the leader, examining the boundary-spanning capacity. The results suggest that different positioning of engineering team leaders in these networks can enhance or impede creativity.

As the analyses show, the relationship between leader centrality in the workflow network and creativity is inversely U-shaped. Translated into managerial terms, this means that leaders who are strongly or minimally involved in the workflow processes negatively affect the creativity of their group. There may be several explanations for this finding. When leaders are strongly involved in the workflow they may easily become overloaded with knowledge, information, and decision needs. This, in turn, leads to two effects that negatively impact on creativity. First, the responsibilities become centralized, discouraging team members, and the team leader becomes a hampering bottleneck when it comes to providing knowledge and information. On the other hand, when team leaders are minimally or not at all involved in workflow processes, they cannot monitor communication problems or situations in which the team has not properly adapted to the tasks at hand.

The centrality of the team leader in the problem-solving network negatively relates to creativity. This result suggests that leaders who become involved in the problem-solving process of the engineering design team are a disturbing factor that impedes creativity. Engineering team members seemingly need autonomy, their own responsibilities, and individual freedom in order to find novel and useful solutions. Research also shows that the overall problem-solving communication of a team can negatively impact on creativity. Certainly communication between team members is essential for exchanging the knowledge and information required, but frequent problem-solving communication

on a daily basis was found to impede creativity (Kratzer et al., 2004). Research in brainstorming groups confirms this result; people create more and more useful ideas about problems when they are working alone rather than in a group (Nijstad, 2000). In addition, in brainstorming groups communication-dominating individuals are also found to negatively influence the search for ideas (Nijstad, 2000). Conversely, leaders who do not dominate the problem-solving process are stimulators of creativity.

The relationship between centrality in the awareness network and creativity is inversely U-shaped. This result is in contrast to the hypothesized positive effect, however has to be treated with caution, because the result is only significant at the 10% level. It would seem that leaders who are too strongly involved in a network containing knowledge about others impede the creative process of engineering design teams. This result may be explained by the preceding arguments. It may be that team members, when seeking the right information and knowledge at the right time, also need autonomy to some degree, as well as the responsibility to act on their own. The creative process involved in searching for information and knowledge perhaps generates the framework and general ideas about how to solve problems, since the searching process involves the framing of the problem at hand.

In other words, before team members can search for certain information and knowledge they have to decide what and where to search. On the other hand, the results also indicate that team leaders should be involved in this process to some degree. There are situations where the search process of team members may not be successful or result in insufficient information and knowledge. In such a situation team leaders can intervene and link the information searcher to the right information provider. In addition, when team leaders are involved in the awareness network to some degree, they are able to monitor any possible problems that may occur in multidisciplinary teams. Alternatively, team leaders may have selected their members on the base of knowing their capabilities and, therefore, require utilizing this network less frequently. The overlap between the 'awareness network' and the other networks may also lessen the costs of searching for capabilities throughout time.

The analysis of the relationship between the boundary-spanning capacity of the team leader and creativity is positive as was expected in the hypothesis. This result confirms earlier research on this topic (Allen, 1977; Hansen, 2002; Szulanski, 1996; Tsai, 2001). Seemingly, team leaders who have stable and continued contacts with other organizational units are better able to provide their teams with the required input. This result is not surprising, since in highly specialized and knowledge-intensive work environments the required information and knowledge is usually distributed across several units. The result also indicates that team leaders have to provide certain communication skills in order to understand, select, and translate the information and knowledge at hand for their own team.

One additional result is that the density of the problem-solving network and the awareness network negatively impacts on the creativity of engineering design teams. This result confirms earlier work on the impact of the density of communication networks on team creativity in an innovative setting (e.g., Kratzer et al., 2004). As this research shows, the results do not suggest that not communicating at all contributes to creativity, but that there is a certain threshold of communication density at which any more communication hinders creative achievement (Leenders et al., 2003). There are other factors that also influence the creativity of engineering design teams. One factor that is frequently mentioned is the friendship ties between the team members. Friendship may ease the flow of information and knowledge within engineering design teams by establishing better understanding, openness and trust among the team members. In an earlier study we investigated the effect of friendship ties on team creativity and found that friendship indeed related positively to team creativity (Kratzer et al., 2005). In the current study, however, we focused our interest on the work-related aspects of communication.

To summarize our study, the results confirm that leadership plays an important role in understanding the secrets of creativity. The position of the team leader in the engineering design team promotes creativity when it is moderately centralized in the workflow network, decentralized in the problem-solving network, moderately centralized in the awareness network and very central in the external information network.

7. Conclusions and directions for future research

The postulate that not only personal characteristics, but also environmental factors would impact on the creativity of teams has been confirmed, and so this is one empirical finding made richer by this

study. It can be concluded that different network positions in different networks of team leaders affect the creativity of engineering design teams. The results show positive, inversely U-shaped and negative relationships. In this analyses it has turned out that degree centrality is a powerful measure when studying leadership. In addition, the three communication networks studied are distinguishable by the flows of information and knowledge in their engineering design teams. These findings allow for two additional conclusions to be drawn.

First, the analysis of social networks is a powerful tool in explaining environmental effects on creativity. Second, there are different flows of information and knowledge in engineering design teams, and each flow has its own characteristic and effect on creativity. In addition, the choice made to investigate the three distinctive networks increases the validity of the results found. In managerial terms one implicit conclusion may be drawn. The study indicates that it is not the leaders strongly involved in internal networks that stimulate teams towards creativity, but rather the leaders who are able to gather the necessary information and knowledge outside the team and to distribute this intelligently within the team. The choice to promote team members to team leaders, therefore, should be based on managerial capabilities rather than on professional expertise.

Since our study has its limitations, research in investigating the relationship between leadership and creativity should continue, using the explanatory power of social networks, distinguishing among different flows of information and knowledge, and testing also different measures of creativity based on more objective ratings. Moreover, creativity is not required in all situations. There are engineering design teams who face more routine tasks. In the literature it is well documented that different structural configurations may serve different goals. Since we are especially interested in the structural position of leadership and creativity, the results cannot readily be translated into situations in which teams face more routine tasks where aspects of performance other than creativity are more in demand, such as, for example, effectiveness, reliability, or productivity.

Research, therefore, should also investigate the structural position of leadership and other aspects of performance. Moreover, future research should concentrate on larger datasets that allow for the inclusion of more variables in the analyses, such as different branches in which engineering design teams are engaged, company sizes, etc. We went in this direction in part when we included another large instrument consortium in our research. A second suggestion for future research would be to study data in a longitudinal setting. Social networks are dynamic systems that may change over time, in particular when a social network has just been established. Moreover, the very process of research and development itself refers to a changing structure of tasks and responsibilities over time. The networks in the study we have presented had a tenure of six to seven years, so that one should be able to expect them to be stable. Still, a more longitudinal setting would permit investigation of how engineering design teams are established and how the dynamic path works through the innovation cycle, and so subsequently causal conclusions would be drawn in a much stronger way.

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